1.2 Algebra & Functions

Mathematics — OCR A-Level (H240) — Paper 1: Pure Mathematics

1. Indices and Surds

The rules of indices (exponents) underpin nearly every algebraic manipulation in the A-Level course. They hold for any real base $a > 0$ and any real exponents m, n; for integer exponents they hold for any non-zero base.

Laws of Indices

Law	Statement	Example
Product	$a^m \cdot a^n = a^{m+n}$	$x^3 \cdot x^5 = x^8$
Quotient	$a^m / a^n = a^{m-n}$	$x^7 / x^2 = x^5$
Power of a power	$(a^m)^n = a^{mn}$	$(x^4)^3 = x^{12}$
Power of a product	$(ab)^n = a^n b^n$	$(2x)^3 = 8x^3$
Power of a quotient	$(a/b)^n = a^n / b^n$	$(x/3)^2 = x^2/9$
Zero index	$a^0 = 1$ (for $a \neq 0$)	$7^0 = 1$
Negative index	$a^{-n} = 1/a^n$	$x^{-3} = 1/x^3$
Fractional index	$a^{1/n} = \sqrt[n]{a}$, $a^{m/n} = \sqrt[n]{a^m}$	$8^{2/3} = (\sqrt[3]{8})^2 = 4$

Surd. An irrational number expressed in the form $\sqrt[n]{a}$ where a is a positive integer and the n-th root

is not itself a rational number. Common surds at A-level are $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{10}$.

Simplifying Surds

Key rules: $\sqrt{ab} = \sqrt{a}\sqrt{b}$ and $\sqrt{a/b} = \sqrt{a}/\sqrt{b}$, valid for $a, b \geq 0$. To simplify $\sqrt{N}$, factor out the largest perfect-square divisor of N.

Worked Example 1 — Simplifying surds

Simplify (a) $\sqrt{50}$, (b) $\sqrt{72} - \sqrt{18}$, (c) $\sqrt{12} \cdot \sqrt{27}$.

```
// (a) factor out the largest square
1.  √50 = √(25 · 2) = √25 · √2 = 5√2.

// (b) simplify each surd then combine
2.  √72 = √(36 · 2) = 6√2
3.  √18 = √(9 · 2)  = 3√2
4.  √72 − √18 = 6√2 − 3√2 = 3√2.

// (c) combine under one radical
5.  √12 · √27 = √(12 · 27) = √324 = 18.
```

Rationalising the Denominator

A denominator containing a surd is "rationalised" by multiplying numerator and denominator by a convenient surd expression to remove the irrational part.

- $\dfrac{1}{\sqrt{a}} = \dfrac{\sqrt{a}}{a}$ — multiply top and bottom by $\sqrt{a}$.
- $\dfrac{1}{a+\sqrt{b}} = \dfrac{a-\sqrt{b}}{a^2-b}$ — multiply by the *conjugate* $a - \sqrt{b}$.
- $\dfrac{1}{\sqrt{a}+\sqrt{b}} = \dfrac{\sqrt{a}-\sqrt{b}}{a-b}$.

Worked Example 2 — Rationalising

Express $\dfrac{5}{3-\sqrt{2}}$ in the form $p+q\sqrt{2}$.

```
1. Multiply by conjugate (3 + √2)/(3 + √2):
2.   5/(3 − √2) · (3 + √2)/(3 + √2)
3. = 5(3 + √2) / (3² − (√2)²)
4. = (15 + 5√2) / (9 − 2)
5. = (15 + 5√2) / 7
6. = 15/7 + (5/7)√2.     // so p = 15/7, q = 5/7
```

Exam tip. OCR examiners routinely deduct marks when students leave a surd in the denominator. Always rationalise as the final step unless the question explicitly leaves an alternative form. If the denominator is $a \pm \sqrt{b}$, multiply by $a \mp \sqrt{b}$ — the difference of two squares does the work for you.

2. Quadratics

A quadratic has the form ax^2+bx+c with $a \neq 0$. Three standard techniques solve $ax^2+bx+c=0$: factorising, completing the square, and the quadratic formula.

Quadratic formula. The solutions of $ax^2+bx+c=0$ (with $a \neq 0$) are $x = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a}$.

Discriminant. The quantity $\Delta = b^2-4ac$. Its sign determines the number and type of real roots and the way the parabola meets the x-axis.

Discriminant Cases

Sign of Δ	Roots	Graph behaviour
$\Delta > 0$	Two distinct real roots	Parabola crosses x-axis at two points
$\Delta = 0$	One repeated real root	Parabola touches x-axis (tangent) at vertex

$\Delta < 0$	No real roots (complex conjugate pair)	Parabola does not meet x-axis
Δ is a perfect square	Rational roots — factorises over $\mathbb{Q}$	Easiest case to factor by inspection

Completing the Square

Any quadratic $ax^2 + bx + c$ can be written in *vertex form*: $a(x-h)^2 + k$, where $h = -\dfrac{b}{2a}$ and $k = c - \dfrac{b^2}{4a}$. The vertex of $y = ax^2 + bx + c$ sits at (h, k); the minimum (if $a > 0$) or maximum (if $a < 0$) value is k.

Worked Example 3 — Solve $2x^2 - 5x + 3 = 0$ three ways

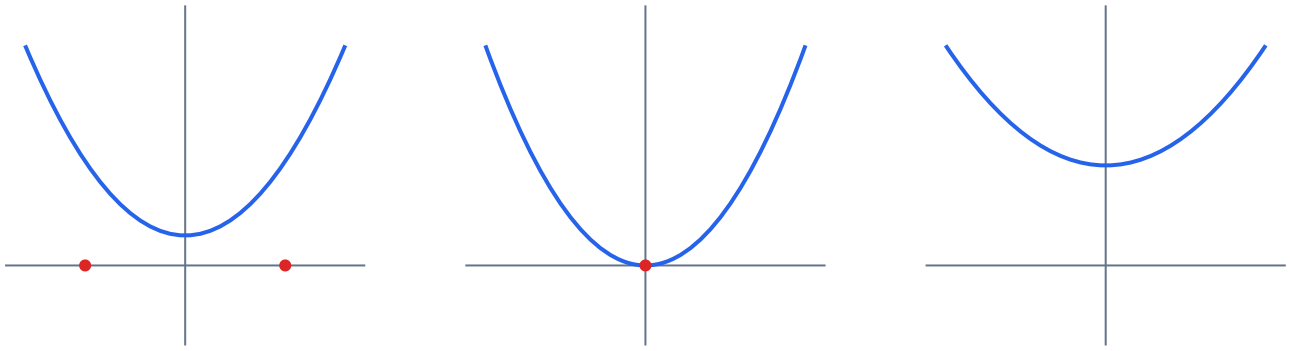
```
// Method A – factorising
 1.  2x² – 5x + 3 = 0
 2.  Find two numbers that multiply to 2·3 = 6 and sum to –5:   –2 and –3.
 3.  Split middle term:   2x² – 2x – 3x + 3 = 0
 4.                        2x(x – 1) – 3(x – 1) = 0
 5.                        (2x – 3)(x – 1)     = 0
 6.  ⇒ x = 3/2  or  x = 1.

// Method B – quadratic formula
 7.  a = 2, b = –5, c = 3, Δ = 25 – 24 = 1.
 8.  x = (5 ± √1) / 4  =  (5 ± 1)/4.
 9.  ⇒ x = 6/4 = 3/2,  or  x = 4/4 = 1.

// Method C – completing the square
10.  2x² – 5x + 3 = 2(x² – 5x/2) + 3
11.               = 2[(x – 5/4)² – 25/16] + 3
12.               = 2(x – 5/4)² – 25/8 + 3
13.               = 2(x – 5/4)² – 1/8.
14.  Set to 0:     2(x – 5/4)² = 1/8
15.                (x – 5/4)²  = 1/16
16.                 x – 5/4    = ±1/4
17.  ⇒ x = 5/4 + 1/4 = 3/2,  or  x = 5/4 – 1/4 = 1.
```

The vertex form also tells us the minimum value of the quadratic is $-1/8$, attained at $x = 5/4$.

Exam tip — when to use which method. Try factorising first: it is fastest when Δ is a small perfect square. If factorising is not obvious in ~15 seconds, switch to the formula. Use completing the square when the question asks for the vertex, the minimum/maximum value, the range, or to prove that the quadratic is always positive/negative.

Sketching Quadratics

To sketch $y = ax^2 + bx + c$: (i) direction of opening — up if $a > 0$, down if $a < 0$; (ii) y-intercept $(0, c)$; (iii) x-intercepts from the roots (if real); (iv) vertex from completing the square.



Δ > 0 Δ = 0 Δ < 0

Worked Example 4 — Quadratic word problem

A rectangular field is 5 m longer than it is wide and has area 84 m². Find its dimensions.

Let width = x. Then length = $x + 5$ and $x(x + 5) = 84$, i.e. $x^2 + 5x - 84 = 0$.

Discriminant $= 25 + 336 = 361 = 19^2$. Roots: $x = (-5 \pm 19)/2$, giving $x = 7$ or $x = -12$. Width must be positive, so $x = 7$ m, length = 12 m.

3. Simultaneous Equations

Simultaneous equations have a solution set consisting of every pair (or n-tuple) that satisfies both (all) equations. Geometrically each equation is a curve; solutions are intersection points.

Linear / Linear Systems

Two methods: **substitution** (solve one equation for a variable, substitute into the other) and **elimination** (scale the equations and add/subtract to cancel a variable). Exactly one solution iff the two lines are not parallel.

Worked Example 5 — Linear/linear

Solve $3x + 2y = 16$ and $x - y = 2$.

```
1.  From (2):  x = y + 2.
2.  Substitute:  3(y + 2) + 2y = 16
3.              3y + 6 + 2y   = 16
4.              5y            = 10
5.              y             = 2,  x = 4.
6.  Check:  3(4) + 2(2) = 16 ✓   and   4 - 2 = 2 ✓.
```

Geometrically, the lines $3x + 2y = 16$ and $x - y = 2$ intersect at $(4, 2)$.

Linear / Quadratic Systems

Substitute the linear equation into the quadratic to reduce to a single quadratic in one variable. Possibilities: two solutions (line is a secant — cuts the curve), one solution (line is tangent), no solutions (line misses the curve). The discriminant of the resulting quadratic tells you which case you are in.

Worked Example 6 — Linear/quadratic

Solve $y = x^2 - 2x - 3$ and $y = x - 3$. Interpret the result geometrically.

```
1.  x² - 2x - 3 = x - 3
2.  x² - 3x     = 0
3.  x(x - 3)    = 0
4.  x = 0  or  x = 3.
```

```
5.  Corresponding y:  y = 0 - 3 = -3;  y = 3 - 3 = 0.
6.  ∴ intersection points: (0, -3) and (3, 0).
```

The line $y = x - 3$ crosses the parabola $y = x^2 - 2x - 3$ at exactly these two points (two real solutions ⇒ line is a secant).

4. Inequalities

When manipulating inequalities, remember: multiplying or dividing both sides by a *negative* number **reverses** the inequality.

Linear Inequalities

Treat like a linear equation, but reverse the sign when multiplying/dividing by negatives.

Example: $3 - 2x > 7 \Rightarrow -2x > 4 \Rightarrow x < -2$.

Quadratic Inequalities

Solve $ax^2 + bx + c > 0$ (or $<, \leq, \geq$) in three steps:

1. Find the roots α, β of $ax^2 + bx + c = 0$.
2. Sketch the parabola (or draw a sign diagram) noting whether it opens up ($a > 0$) or down ($a < 0$).
3. Read off the x-values where the sketch lies above/below the x-axis as required.

Worked Example 7 — Quadratic inequality

Solve $x^2 - x - 6 \leq 0$.

```
1.  Roots of x² - x - 6 = 0:  (x - 3)(x + 2) = 0  ⇒  x = -2, x = 3.
2.  Parabola opens upwards (coefficient of x² is +1).
3.  y ≤ 0 between the roots.
4.  ∴ solution:  -2 ≤ x ≤ 3.
```

Exam tip — sign diagram for a quadratic inequality. Always sketch the parabola, even a rough one. A common error is to write $x < -2$ or $x > 3$ for $x^2 - x - 6 \leq 0$ — that is the solution to the > 0 version. The sketch prevents this inversion of the answer. Remember: *between the roots* for ≤ 0 (upward parabola), *outside the roots* for ≥ 0.

Modulus Inequalities

The modulus $|x|$ means the non-negative value of x. Useful equivalences:

- $|x| = a$ ($a \geq 0$) iff $x = a$ or $x = -a$.
- $|x| \leq a$ iff $-a \leq x \leq a$.
- $|x| \geq a$ iff $x \leq -a$ or $x \geq a$.

Example: $|2x - 1| < 5 \Rightarrow -5 < 2x - 1 < 5 \Rightarrow -4 < 2x < 6 \Rightarrow -2 < x < 3$.

5. Polynomials

Polynomial. An expression of the form $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, where $a_0, a_1, \ldots, a_n$ are real constants with $a_n \neq 0$. The *degree* is n and the *leading coefficient* is a_n.

Polynomial Division

To divide polynomial $f(x)$ by $g(x)$ ($\deg g \leq \deg f$), use long division:

- Divide leading term of f by leading term of g — this is the next term of the quotient.
- Multiply $g(x)$ by that term, subtract from $f(x)$.
- Repeat on the remainder until the remainder has degree $< \deg g$.

Worked Example 8 — Polynomial long division

Divide $2x^3 - 3x^2 - 8x + 12$ by $x - 2$.

```
                2x²  -  x  - 10          // quotient
        ____________________________________
```

```
x − 2 )  2x³ − 3x² − 8x + 12
         2x³ − 4x²                    // subtract:  2x²·(x−2)
        ___________
               x² − 8x
               x² − 2x                // subtract:  x·(x−2)
             _________
                  −10x + 12
                  −10x + 20           // subtract:  −10·(x−2)
                 _________
                       −8             // remainder
```

So $2x^3 - 3x^2 - 8x + 12 = (x-2)(2x^2 - x - 10) - 8$.

Factor and Remainder Theorems

Factor theorem. For a polynomial $f(x)$ and a constant a: $(x-a)$ is a factor of $f(x)$ if and only if $f(a) = 0$.

Remainder theorem. When $f(x)$ is divided by $(x-a)$, the remainder equals $f(a)$. (The factor theorem is the special case where this remainder is 0.)

Worked Example 9 — Factor theorem applied

Factorise $x^3 + 2x^2 - 5x - 6$ completely.

```
1.  Try small integers dividing the constant term −6:  ±1, ±2, ±3, ±6.
2.  f(−1) = −1 + 2 + 5 − 6 = 0.                    // so (x + 1) is a factor
3.  Divide x³ + 2x² − 5x − 6  by  (x + 1):
       quotient  =  x² + x − 6.
4.  Factor the quadratic:  x² + x − 6 = (x + 3)(x − 2).
5.  ∴ x³ + 2x² − 5x − 6 = (x + 1)(x + 3)(x − 2).
6.  Roots:  x = −3, −1, 2.
```

Exam tip — searching for roots. For a polynomial with integer coefficients and leading coefficient 1, any rational root must be an integer dividing the constant term. For leading coefficient a_n, any rational root p/q satisfies $p \mid a_0$ and $q \mid a_n$. Start with ± 1, then ± 2, before trying larger divisors.

Sketching Cubics and Quartics

Given a cubic in factorised form $y = a(x - r_1)(x - r_2)(x - r_3)$:

- Three distinct real roots — the curve crosses the x-axis three times.
- A repeated root $(x - r)^2$ — the curve *touches* (is tangent to) the x-axis at $x = r$.
- A triple root $(x - r)^3$ — the curve has a point of inflection on the x-axis at $x = r$.
- As $x \to +\infty$: $y \to +\infty$ if $a > 0$; $y \to -\infty$ if $a < 0$. Reverse for $x \to -\infty$.

Analogous rules apply to quartics: leading coefficient positive gives "up then up" end behaviour; leading coefficient negative gives "down then down".

6. Graphs of Functions — Transformations

Starting from the graph of $y = f(x)$, a sequence of standard transformations produces related graphs. The four families are *translation*, *stretch*, *reflection*, and combinations thereof.

Transformation Summary

New function	Effect on graph of $y = f(x)$	Worded description
$f(x) + a$	Translate vertically	Shift up by a (down if $a < 0$)
$f(x - a)$	Translate horizontally	Shift right by a (left if $a < 0$)
$a \cdot f(x)$	Vertical stretch	Scale y-coordinates by factor a
$f(ax)$	Horizontal stretch	Scale x-coordinates by factor $1/a$
$-f(x)$	Reflect in x-axis	Flip vertically
$f(-x)$	Reflect in y-axis	Flip horizontally

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$\lvert f(x)\rvert$	Reflect negative parts upwards	Any part below the x-axis is flipped above
$f(\lvert x\rvert)$	Right half mirrored to left	Keep $x \geq 0$ part; reflect it in the y-axis

Exam tip — direction of horizontal shifts. Horizontal transformations behave *opposite* to what the sign suggests. $f(x-3)$ shifts the graph **right** by 3 (because the x-value that used to give $f(0)$ now needs $x = 3$). Similarly $f(2x)$ **compresses** horizontally by factor 2; $f(x/2)$ stretches by factor 2. Many students get the sign wrong under pressure — practice this until it's automatic.

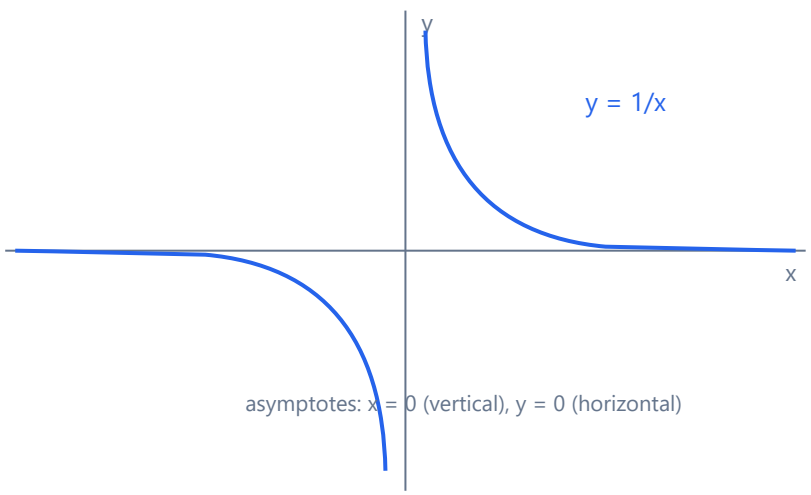
Combining Transformations

Order matters. For $y = af(b(x-c)) + d$: the inside ($b(x-c)$) acts first on x — horizontal stretch by $1/b$, then translation right by c; the outside ($a \cdot +d$) acts next on y — vertical stretch by a, then translation up by d.

7. Rational Functions and Asymptotes

A rational function is a quotient of two polynomials, $y = P(x)/Q(x)$. The standard A-level example is $y = 1/x$.

- **Vertical asymptote**: $x = a$ where $Q(a) = 0$ but $P(a) \neq 0$. The function tends to $\pm\infty$ as $x \to a$.
- **Horizontal asymptote**: behaviour as $x \to \pm\infty$. If $\deg P < \deg Q$, the horizontal asymptote is $y = 0$; if $\deg P = \deg Q$, it is y = ratio of leading coefficients.



Translations apply as usual: $y = 1/(x-1)$ shifts the graph of $y = 1/x$ right by 1, giving a vertical asymptote at $x = 1$. $y = 1/x + 2$ shifts it up by 2, giving a horizontal asymptote at $y = 2$.

8. Partial Fractions

Partial fractions. A technique that decomposes a proper rational expression $P(x)/Q(x)$ (with $\deg P < \deg Q$) into a sum of simpler fractions whose denominators are the factors of $Q(x)$.

If $\deg P \geq \deg Q$, first perform polynomial division to obtain a polynomial plus a proper rational remainder, then decompose the remainder.

Setup by Denominator Type

Factor in $Q(x)$	Corresponding partial fraction(s)
Distinct linear $(x-a)$	$\dfrac{A}{x-a}$
Repeated linear $(x-a)^2$	$\dfrac{A}{x-a} + \dfrac{B}{(x-a)^2}$
Repeated linear $(x-a)^3$	$\dfrac{A}{x-a} + \dfrac{B}{(x-a)^2} + \dfrac{C}{(x-a)^3}$
Irreducible quadratic (x^2+bx+c), $\Delta < 0$	$\dfrac{Ax+B}{x^2+bx+c}$

Worked Example 10 — Distinct linear factors

Express $\dfrac{3x}{(x-1)(x+2)}$ in partial fractions.

```
Set up:   3x / [(x - 1)(x + 2)]  =  A/(x - 1) + B/(x + 2).
Multiply through by (x - 1)(x + 2):
              3x = A(x + 2) + B(x - 1).
Substitute x = 1:   3(1)  = A(3) + B(0)   ⇒   A = 1.
Substitute x = -2:  3(-2) = A(0) + B(-3)  ⇒   B = 2.
```

```
6.  ∴  3x / [(x − 1)(x + 2)]  =  1/(x − 1) + 2/(x + 2).
7.  Check:  1/(x−1) + 2/(x+2) = [(x+2) + 2(x−1)] / [(x−1)(x+2)] = 3x / [(x−1)(x+2
```

Worked Example 11 — Repeated linear factor

Express $\dfrac{2x+1}{(x-1)(x+2)^2}$ in partial fractions.

```
1.  Set up:  (2x + 1) / [(x − 1)(x + 2)²]
           = A/(x − 1) + B/(x + 2) + C/(x + 2)².
2.  Multiply up:   2x + 1 = A(x + 2)² + B(x − 1)(x + 2) + C(x − 1).
3.  x = 1:    3   = A(9) + 0 + 0            ⇒   A = 1/3.
4.  x = −2:  −3   = 0 + 0 + C(−3)          ⇒   C = 1.
5.  Compare x² coefficients: 0 = A + B       ⇒   B = −1/3.
6.  ∴ (2x+1)/[(x−1)(x+2)²]
       = 1/[3(x − 1)] − 1/[3(x + 2)] + 1/(x + 2)².
```

Exam tip — setting up partial fractions. The most common error is forgetting the repeated-factor rule. $(x-a)^2$ contributes *two* terms: $A/(x-a)$ and $B/(x-a)^2$. Missing the first of these produces a system with no solution — if you find your simultaneous equations are inconsistent, check the setup before hunting for arithmetic slips.

9. Functions

Function. A rule $f : A \to B$ that assigns to each element x of a set A (the *domain*) exactly one element $f(x)$ of B. The set of actual output values $\{f(x) : x \in A\}$ is the *range* (a subset of B, the codomain).

Key Terminology

- **One-to-one (injective).** Different inputs give different outputs: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Graph passes the *horizontal line test* — every horizontal line meets the graph in at most one point.

- **Many-to-one.** Two or more inputs map to the same output. E.g. $f(x) = x^2$ on $\mathbb{R}$ is many-to-one since $f(2) = f(-2) = 4$.
- **Onto (surjective).** Every element of the codomain is hit: range = codomain.
- **Bijective.** Both injective and surjective — there is a perfect pairing between domain and codomain. A function is invertible if and only if it is bijective (or, by restricting the codomain to the range, merely injective).

Inverse function. If $f : A \to B$ is one-to-one, its inverse $f^{-1} : \mathrm{Range}(f) \to A$ is defined by $f^{-1}(y) = x$ iff $f(x) = y$. The graph of $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in the line $y = x$. Domain of f^{-1} = Range of f; Range of f^{-1} = Domain of f.

Finding an Inverse Algebraically

1. Write $y = f(x)$.
2. Swap x and y (or rearrange to make x the subject, then relabel).
3. Solve for the new y; this gives $f^{-1}(x)$.
4. State the domain of f^{-1} — it is the range of f.

Worked Example 12 — Inverse of a rational function

Let $f(x) = \dfrac{2x+3}{x-1}$, defined for $x \in \mathbb{R}$ with $x \neq 1$. Find $f^{-1}(x)$ and state its domain.

```
Let y = (2x + 3)/(x − 1).
Multiply up:        y(x − 1) = 2x + 3
                     yx − y   = 2x + 3
                     yx − 2x  = y + 3
                     x(y − 2) = y + 3
                     x        = (y + 3)/(y − 2).
Swap variables:     f⁻¹(x)   = (x + 3)/(x − 2).
Domain of f⁻¹ = Range of f.
  Since f(x) = 2 + 5/(x − 1), f takes every real value except 2.
∴ Domain of f⁻¹ is  { x ∈ ℝ : x ≠ 2 }.
```

Note the domain of f^{-1} is *not* simply "all reals" — the value $x = 2$ makes the denominator zero. Stating the domain is expected on every inverse-function question.

Exam tip — always state the domain. OCR mark schemes routinely allocate a dedicated mark for the domain of f^{-1}. The domain of f^{-1} is the range of f; find it by (i) sketching $y = f(x)$, (ii) completing the square, or (iii) finding values f cannot attain. Forgetting this costs a mark almost every time.

Composition of Functions

The composition $(f \circ g)(x) = f(g(x))$ means: apply g first, then apply f to the result. In general $f \circ g \neq g \circ f$. The domain of $f \circ g$ is $\{x \in \mathrm{Domain}(g) : g(x) \in \mathrm{Domain}(f)\}$.

Example: if $f(x) = x + 2$ and $g(x) = x^2$, then $(f \circ g)(x) = x^2 + 2$ whereas $(g \circ f)(x) = (x + 2)^2 = x^2 + 4x + 4$.

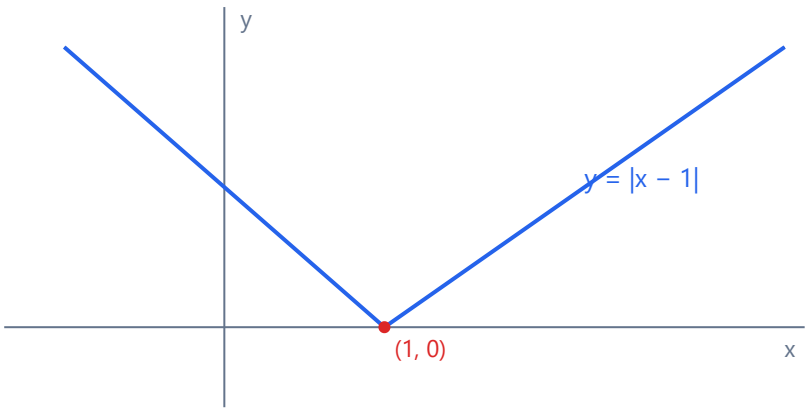
Self-inverse property: $f \circ f^{-1}(x) = x$ and $f^{-1} \circ f(x) = x$ on the appropriate domains. This is often used as a verification step.

10. The Modulus Function

Modulus (absolute value). $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$. Equivalently $|x| = \sqrt{x^2}$. Geometrically, $|x|$ is the distance from x to 0 on the number line; $|x - a|$ is the distance from x to a .

Graph of $y = |f(x)|$

Sketch $y = f(x)$, then reflect any part below the x-axis in the x-axis.



Solving Modulus Equations

Two standard approaches to $|f(x)| = g(x)$:

- **Case method.** $|f(x)| = g(x) \Leftrightarrow f(x) = g(x)$ *or* $f(x) = -g(x)$, with a final check that $g(x) \geq 0$ at each candidate.
- **Graphical method.** Sketch both curves; intersection points are the solutions.

Worked Example 13 — Modulus equation

Solve $|2x - 3| = x + 4$.

```
// Case 1: 2x − 3 ≥ 0, i.e. x ≥ 3/2
1.   2x − 3 = x + 4  ⇒  x = 7.  Check x ≥ 3/2 ✓ and RHS = 11 > 0 ✓.

// Case 2: 2x − 3 < 0, i.e. x < 3/2
2.   −(2x − 3) = x + 4  ⇒  −2x + 3 = x + 4  ⇒  −3x = 1  ⇒  x = −1/3.
3.   Check: x < 3/2 ✓, and RHS = −1/3 + 4 = 11/3 > 0 ✓.

4.   ∴ solutions:  x = −1/3  or  x = 7.
```

For inequalities $|f(x)| \leq g(x)$ equivalent to $-g(x) \leq f(x) \leq g(x)$ (when $g(x) \geq 0$). For $|f(x)| \geq g(x)$, the solution is $f(x) \leq -g(x)$ or $f(x) \geq g(x)$.

11. Curve Sketching — Putting It All Together

A complete sketch of $y = f(x)$ should show:

1. **Intercepts** — values where $x = 0$ (the y-intercept) and $f(x) = 0$ (the x-intercepts).
2. **Stationary points** — solve $f'(x) = 0$; classify as max/min/inflection.
3. **Asymptotes** — vertical where the function is undefined, horizontal from end behaviour.
4. **End behaviour** — as $x \to \pm\infty$, what does $f(x)$ do?
5. **Symmetries** — is f even ($f(-x) = f(x)$) or odd ($f(-x) = -f(x)$)?

Worked Example 14 — Sketch $y = x^3 - 3x$

```
1.  Intercepts: y(0) = 0. Roots of x³ − 3x = x(x² − 3) = 0 ⇒ x = 0, ±√3.
2.  Stationary points: y' = 3x² − 3 = 0 ⇒ x = ±1.
3.    At x = 1: y = 1 − 3 = −2 (local min, y'' = 6x = 6 > 0).
4.    At x = −1: y = −1 + 3 = 2 (local max, y'' = −6 < 0).
5.  End behaviour: x → +∞ ⇒ y → +∞;  x → −∞ ⇒ y → −∞.
6.  Symmetry: f(−x) = −x³ + 3x = −f(x) − odd, rotationally symmetric about origin
```

12. Binomial Expansion — Positive Integer Index

Binomial coefficient. For non-negative integers $n \geq r$, $\binom{n}{r} = {}^nC_r = \dfrac{n!}{r!(n-r)!}$. It counts the number of ways of choosing r objects from n, and equals the coefficient of x^r in $(1+x)^n$.

The **Binomial Theorem** (for $n \in \mathbb{Z}^+$):

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \cdots + b^n.$$

Pascal's Triangle

Row n of Pascal's triangle gives $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$. Each entry is the sum of the two entries above it.

```
n = 0:                    1
n = 1:                  1   1
```

```
n = 2:                1   2   1
n = 3:              1   3   3   1
n = 4:            1   4   6   4   1
n = 5:          1   5   10  10   5   1
n = 6:        1   6   15  20  15   6   1
```

Worked Example 15 — Expand $(1+2x)^5$

```
1.  Use (a + b)^5 with a = 1, b = 2x. Coefficients from row 5: 1, 5, 10, 10, 5, 1
2.  (1 + 2x)^5
      = 1 + 5(2x) + 10(2x)² + 10(2x)³ + 5(2x)⁴ + (2x)^5
      = 1 + 10x + 40x² + 80x³ + 80x⁴ + 32x⁵.
3.  Check constant term (x=0): 1 ✓.  Check sum of coefficients (x=1): 3⁵ = 243 =
```

Use in approximation. If x is small, $(1+2x)^5 \approx 1+10x+40x^2$, because higher powers of x are negligible. This lets you approximate numerical values such as 1.02^5 by setting $x=0.01$: $1+10(0.01)+40(0.01)^2 = 1+0.1+0.004 = 1.104$, compared with the exact $1.1040808\ldots$.

13. Binomial Expansion — Rational or Negative Index

For index n *not* a non-negative integer, the expansion is an *infinite series*:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

The series converges (and equals $(1+x)^n$) only when $|x|<1$. For $(a+x)^n$ with $a \neq 1$, factor out a first: $(a+x)^n = a^n\left(1+\frac{x}{a}\right)^n$, valid for $|x/a|<1$, i.e. $|x|<|a|$.

Worked Example 16 — First four terms of $(1+x)^{-2}$

```
1.  n = -2. Terms:
2.   Term 0:  1
3.   Term 1:  n · x              = (-2) x              = -2x
```

```
Term 2:  n(n−1)/2! · x²       = (−2)(−3)/2 · x²     = 3x²
Term 3:  n(n−1)(n−2)/3! · x³  = (−2)(−3)(−4)/6 · x³ = −4x³
∴ (1 + x)^(−2) ≈ 1 − 2x + 3x² − 4x³ + … ,   valid for |x| < 1.
```

Cross-check: differentiate the geometric series $1/(1-u) = 1 + u + u^2 + \cdots$ or recall $1/(1+x)^2 = \sum_{k\geq 0}(-1)^k(k+1)x^k = 1 - 2x + 3x^2 - 4x^3 + \cdots$. ✓

Exam tip — validity. When you write an expansion of $(1+x)^n$ for non-integer n, *always* state the range of validity ($|x| < 1$, or the appropriate scaled version). The "validity" mark is separate from the expansion marks on nearly every OCR paper.

14. Consolidated Summary and Exam Strategy

Chapter-wide checklist for Algebra and Functions.

- **Indices:** convert radicals to fractional powers before manipulating (e.g. $\sqrt[3]{x^2} = x^{2/3}$).
- **Surds:** always rationalise; keep surds simplified ($\sqrt{50} = 5\sqrt{2}$).
- **Quadratics:** factorise first, formula second, completing the square when you need the vertex/range/min-max.
- **Inequalities:** sketch or use a sign diagram; reverse the sign when multiplying/dividing by a negative.
- **Polynomial division / factor theorem:** test $\pm 1, \pm 2, \pm 3$ before larger divisors of the constant term.
- **Transformations:** inside the bracket = horizontal and reversed; outside = vertical and in the obvious direction.
- **Partial fractions:** match the factor structure exactly — repeated factors need multiple terms; irreducible quadratics need a linear numerator.
- **Functions:** always state the domain of an inverse. Check one-to-one before asserting invertibility.
- **Modulus:** two cases, or the graphical method — always verify candidate solutions against the original equation.

- **Binomial:** for integer n, finite expansion via Pascal's triangle; for rational/negative n, infinite series convergent for $|x| < 1$ — state validity.

Typical OCR examiner complaints (from mark reports).

- Leaving surds in denominators when a "simplified" answer is requested.
- Not stating the domain of f^{-1}.
- Omitting the validity $|x| < 1$ after a binomial expansion for non-integer index.
- Reversing the solution of a quadratic inequality (inside-vs-outside the roots confusion).
- Forgetting the repeated-factor term $B/(x-a)^2$ in a partial fraction setup.
- Accepting an extraneous "solution" to a modulus equation without checking that both sides have the same sign.